

OPAL-RT User Manual (Adapted from Arash Omid)

Alec Benedict, Kendall Meienhofer, Janessa Green, Emily Ninestein

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1 Device description

The Opal-RT system consists of an OP5033, which serves as the real-time simulator comprising CPUs, and an OP4610, which includes an FPGA processor and an I/O expansion unit. The OP5033 simulator functions as a master system for the OP4610 FPGA and I/O expansion units.



(a) OP5033 Opal-RT real-time simulator (b) OP4610-FPGA Processor and I/O Expansion Unit

Figure 1: OPAL-RT System

Table 1: Specifications of OP5033

Parameter	Details
Part Number	OP5033XG-4
CPU	Intel® Xeon® 4 cores - 3.80 GHz
Operating System	OPAL-RTLinux 3.x
Memory	32 GB of DRAM
Hard Disk	500 GB SSD disk
PC interfaces	Standard PC connectors (monitor, keyboard, mouse, and network)
PCIe slots	1 PCI-E 3.0 x4
USB	2 USB-A 3.0 Ports (front) and 2 USB-A 2.0 Ports (back)
LAN	2x RJ45 Gigabit Ethernet LAN ports

Table 2: Specifications of OP4610

Parameter	Details
FPGA	Kintex-7 410T
High-Speed Communication	4x Small Form-factor Pluggable (SFP), 1 to 5 Gbits/s optical fiber
Software Platform	RT-LAB and HYPERSIM
32 Digital I/O	32 channels high range digital input output, Digital out: 50 mA per channel, 5-24 V push-pull FET, Digital in: 0-30 V, DIO selectable per group of 8 channels, 32 static digital
2x 16 Analog Output	16 channels analog output, 1 MS/s (16 channels) or 2 MS/s (8 channels), 16-bit resolution, 15 mA, ± 16 V
16 Analog Input	16 channels analog input, 1 MS/s, 16-bit resolution, 500 ns, ± 20 V
Digital I/O RS422	2 channels for encoder or 6 PWM in / 6 PWM out or other applications requiring reading or generation of fast differential logic signals, 5V

2 OP5033 and PC Connection

The OP5033 is connected to the PC via a LAN cable. To establish this connection, the IP settings on the PC must be configured.

1. **Step 1**: Ensure that the LAN connection is properly established between the PC and the OP5033.
2. **Step 2**: Go to *Settings* $\rightarrow$ *Network and Internet* $\rightarrow$ *Ethernet* $\rightarrow$ *Edit IP assignment*.
3. **Step 3**: Change the IP assignment to *Manual*:
 - Turn on *IPv4*.
 - Set the IP address similar to Fig. 2:
 - **IP Address**: 10.64.117.44
 - **Subnet Mask**: 255.255.255.0
 - **Gateway**: 10.64.117.48
 - **Preferred DNS**: 8.8.8.8

Save the settings.

4. **Step 4**: Test the connection:
 - Open *Command Prompt*.
 - Type: 'ping 10.64.117.48'.
 - If you receive 4 replies, the Opal-RT is successfully connected to the PC.

Edit IP settings

Manual

IPv4

☒ On

IP address

10.64.117.44

Subnet mask

255.255.255.0

Gateway

10.64.117.48

Preferred DNS

8.8.8.8

DNS over HTTPS

Off

Alternate DNS

Save Cancel

Figure 2: IP Setting

3 RT-LAB and License Installation

RT-LAB serves as the interface for Opal-RT systems and must be installed according to the available license. The current RT-LAB version is RT-LAB 2024.4. Additionally, ensure compatibility between MATLAB and RT-LAB versions by checking the official Opal-RT [website](#).

RT-LAB installation is straightforward. You just need to install two licenses for the RT-LAB: one on your target simulator OPT907454 (OP5033XG-4) and one on localhost PC3023929994. Note that all our licences are node-locked and can only be installed on the specific system for which they were tailored.

3.1 License Installation on Windows PC (host)

1. Before starting, ensure that you have received the required localhost license. Copy the license file to the host (remember the path on the drive).
2. Install the license: Open the localhost (double click [1]), select the “License” tab [2], and then click “Install” [3] to open a file browser.
3. Browse to the localhost license file and click “Open”. The file will be automatically installed on the host computer.
4. The license view will be updated with the localhost license information.

3.2 License installation on real-time simulator (target)

Install the license: Open the target (double click [1]), select the “License” tab [2] then click “Install” [3] to open a file browser. Browse to the license file and click Open. The file will be installed on the target automatically. The license page will be updated with the new license data. This page contains the description, the license number, the expiration date and the list of available features for this target.

4 Simulink Model for RT-LAB

A Simulink model compatible with RT-LAB consists of two subsystems: **sm-"name"** and **sc-"name"**, as shown in Fig.3. The “sm” subsystem contains the system to be simulated, while the “sc” subsystem includes the “OpComm” block and components for accessing or monitoring the simulation, such as parameter adjustments and scopes, as depicted in Fig.4. **WHEN RUNNING THE SIMULATION ONLY THE SC SUBSYSTEM WILL BE ACCESSIBLE**The outputs from the “sm” subsystem should be connected to the “OpComm” block. Once the model is built, it should be saved in both *.mdl and *.slx formats.

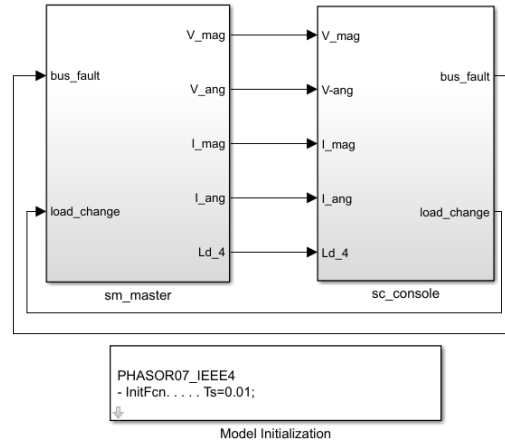


Figure 3: Simulink Model for RT-LAB

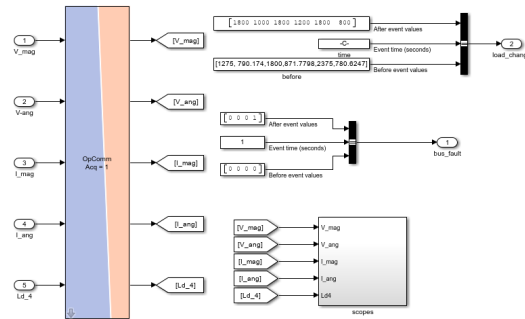


Figure 4: SC subsystem

5 Creating a New Project in RT-LAB

- Open RT-LAB and select a workspace.
- Ensure that both `localhost` and `rtserver` (Figure 5) are activated. Refresh server if necessary (right click then click refresh).

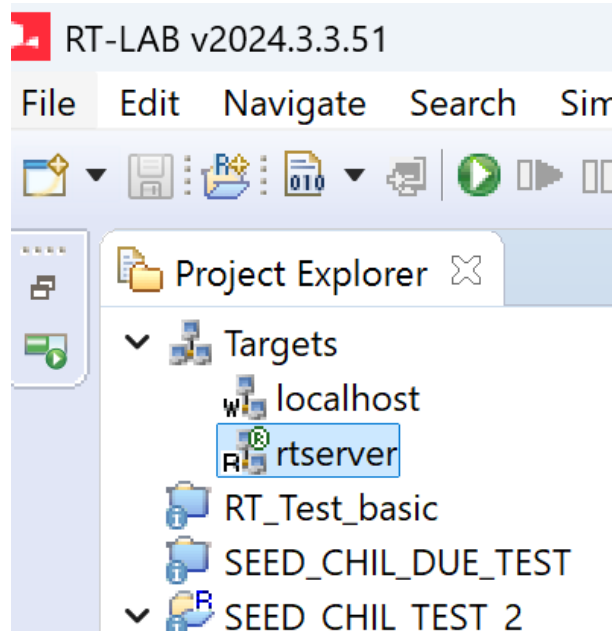


Figure 5: Active localhost/rtserver

- In the “Project Explorer,” click on “Create a new project” (Figure 6).

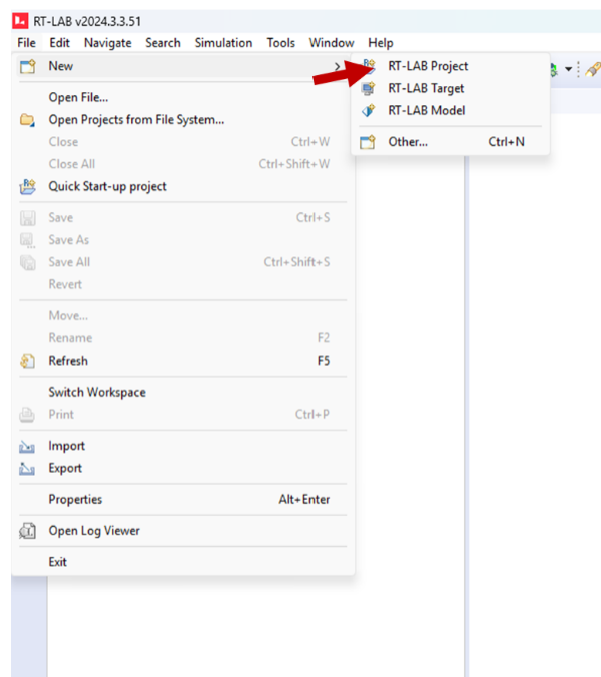


Figure 6: Create New Project

- Enter a name for the project and click “Next.”
- If you want to build your own project, click “Finish.”

- Locate your project in the “Project Explorer” and click on the sidebar.
- Right-click on “Models” in the menu and select “Import then Existing RT-LAB Model” (Figure 7).

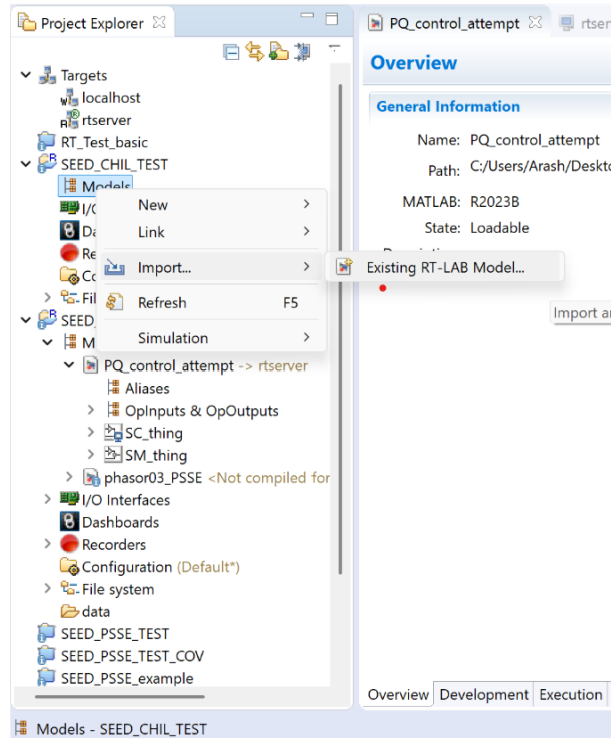


Figure 7: Import Existing Model

- Choose your model file (*.slx).
- In the next window, select both *.mdl (if there is no .mdl save a copy of the .slx as a .mdl) and *.slx files. If there is an associated *.m, *.xlsx, *.raw, *.dyr file, select them as well. Click “Finish.” The file types are explained in Table 3.

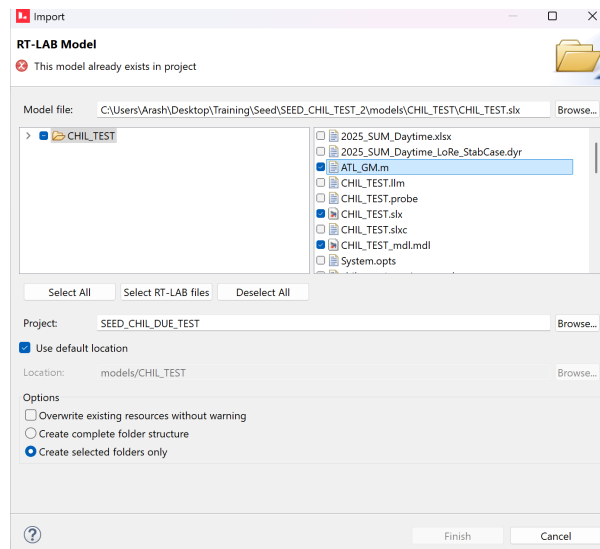


Figure 8: Select .slx/.mdl as well as other project required files

File Type	Explanation
.m	MATLAB file used to initialize SIMULINK variables.
.mdl	Legacy SIMULINK model file (required for RT-LAB).
.slx	SIMULINK file containing the SIMULINK model including RT-LAB and ePHASORSIM blocks.
.raw	Case data file containing the attributes of the PSS/E powerflow case.
.dyr	Dynamic data file containing dynamic behavior models of grid devices.
.xlsx	Excel file containing desired ePHASORSIM input/outputs.

Table 3: RT-Lab Simulation File Types and Explanations

- To edit the model, right-click on it and select “Edit.” MATLAB will open, allowing you to modify the model. **BEFORE PROCEEDING RUN THE SIMULINK MODEL TO VERIFY IT WORKS CORRECTLY.**
- Select “Build.” If the model builds successfully, you will see the message “Completed successfully.” This can take 3-4 minutes.
- Click “Load” to load your model onto OPAL-RT. You should see the “SC Subsystem” components.
- Click “Execute” to run the model.
- click “Pause” to stop the simulation.
- Once the simulation is complete, click “Reset” reset the model.
- The simulation can be re performed by loading/executing again. If you change the model .slx you must re build/load/execute.
- **BEFORE CLOSING RT LAB, CLOSE ALL OPEN PROJECTS (Right click project → close).**

6 Analog and Digital Input/Output

To run Hardware in the Loop Simulation, the OPAL-RT I/O Expansion unit can be used. Both Analog and digital signals can be used. The OP4610 contains 2 DB37 connections for each analog/digital input and output. For more information on the OP4610: [Datasheet](#)

To run HIL the SIMULUNK model must be adapted. For both analog and digital inputs and outputs, the signals must first be connected to the appropriate Simulink blocks, as shown in Fig. 9. To send a signal as an output, use the `OpOutput` block. To receive a signal from OPAL-RT, use the `OpInput` block. To add these blocks:

- In the project workspace click “Edit” to open the Simulink model.
- In the library browser (under the simulation tab) search for `OpInput` or `OpOutput` block.
- Add the necessary blocks to your model. Rename the blocks to correspond to their signals.
- The OPAL-RT can only produce Outputs of $\pm 16V$, so the `OpOutput` signals must be scaled. A gain block must be added before the `OpOutput` to scale the output to $\pm 16V$.
- Once you have scaled the output voltages save and close the model.
- Build your model.
- In the project menu, right-click on “I/O Interfaces.”
- Add New “I/O Interfaces” (Figure 10).
- From the list, select “OPAL-RT Board.” Click Finish then right click on OPAL-RT Board to open (Figure 11).



Figure 9: Input and Output blocks in Simulink

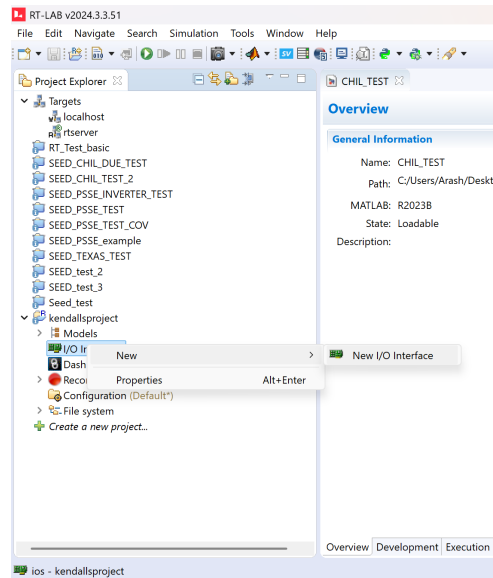


Figure 10: Add I/O Interface

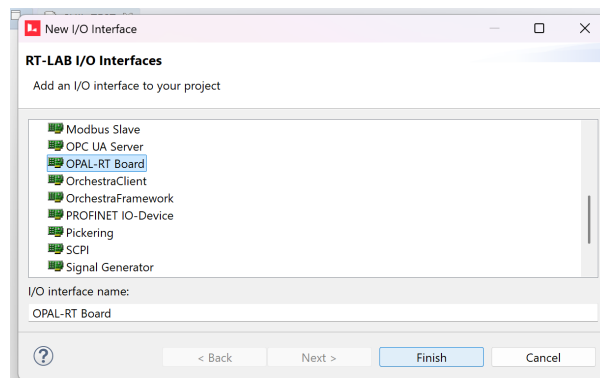


Figure 11: Select OPAL-RT Board

- Click on the “OPAL-RT Board” in the Project Explorer menu.
- Set the “Chassis Type” to OP4610XG (Kintex-7).
- Set the “Chassis ID” to 0.
- In the “Bitstream Configuration” section, select the Bitstream file *.opal. This is found in the OPAL-RT Licenses folder. (Location).

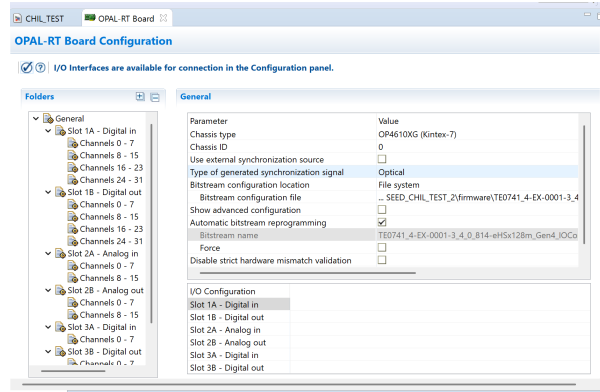


Figure 12: OPAL-RT Board Menu

- In the “Folders” menu, choose the slot number and signal type to open an I/O setup.
- Click on “Channels” based on the number of signals.
- Enable the selected channel. Save (ctrl+s).

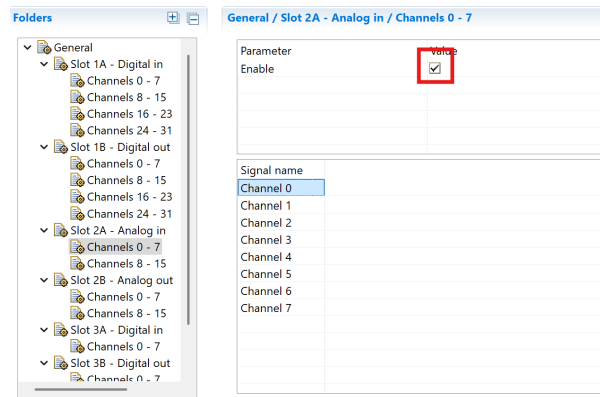


Figure 13: Enable Channel

- In the Project Explorer menu, select “Configuration.”
- Choose the “OPAL-RT Board” to view the slots and channels you have selected.

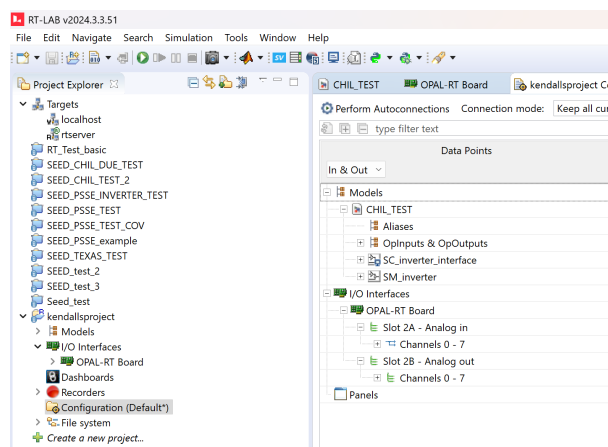


Figure 14: Configuration Menu

- In the Project Explorer, open the “Model” menu and click on the sidebar of “Inputs and Outputs.” You will see all input and output signals of the model. (May need to close/reopen project for these to appear).
- Drag each signal from the project model menu and assign it to the corresponding slot in the configuration menu connections column.

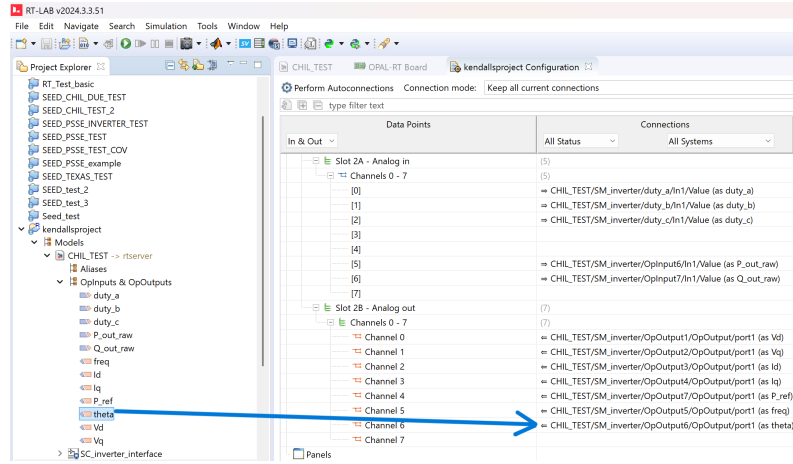


Figure 15: Add I/O To Configuration

- Assign one channel for each input or output.
- Finally, load the model.
- Using a DB37 Cable connect the selected I/O slot/channels to their respective hardware locations.
- Run the simulation according to the instructions in section 5.

7 Import/Export RT-Lab Project

To import and export RT-Lab project follow the [website](#) instructions.

8 Saving Data

When running RT simulation, signals can be logged directly and stored for plotting/analysis. This can be needed as the SIMULINK scopes in the SC subsystem is not meant for plotting real time data and may skip points. To save data **OpWriteFile** blocks can be added to the SIMULINK model and set up according to the instructions below.

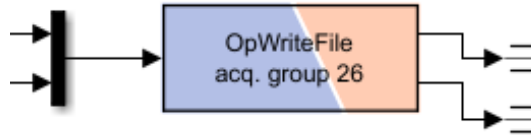


Figure 16: OpWriteFile

- In the RT-LAB project workspace click “edit” to open the SIMULINK model.
- Open the SM subsystem. In the library browser search for **OpWriteFile** and add the block to the model.

- Use a 'Mux' block to collect all signals to be logged and connect the output to the input of the OpWriteFile.
- Connect terminators to the outputs of the OpWriteFile block.
- Click on the OpWriteFile block to open up its mask.

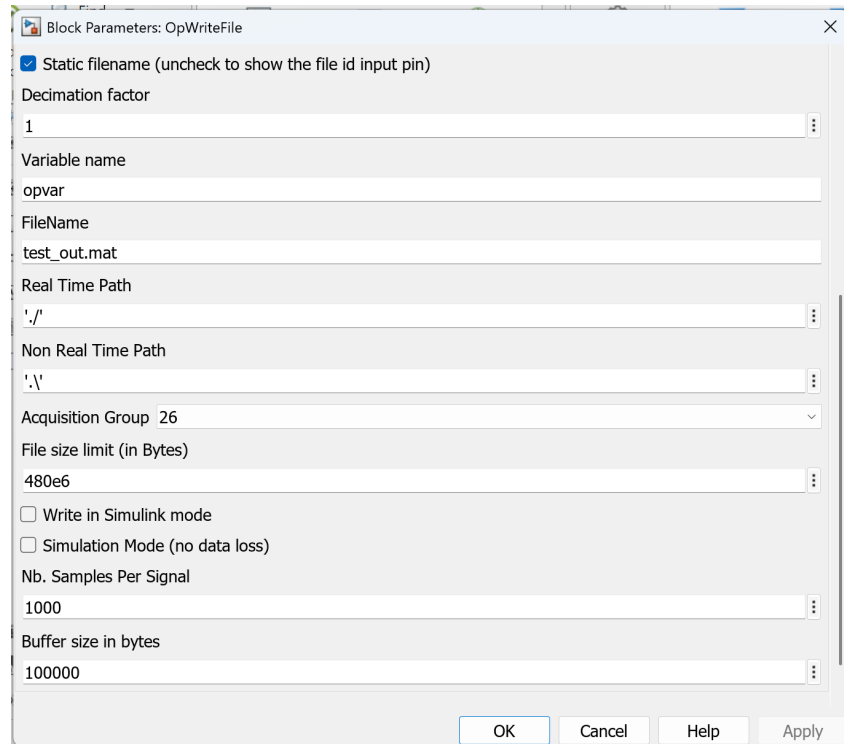


Figure 17: OpWriteFile Mask

- Calculate the file size limit and buffer size according to Figures 19 and 18 Click “Okay” and save the model. *Note: The file can become very large when multiple signals are logged, so be sure to check this before executing the simulation.*

$$\text{Buffer size (bytes)} \geq (\text{Number of logged signals} + 1) \times (\text{Nb. samples}) \times 8$$

Figure 18: Buffer Size Calculation

$$\text{File size (bytes)} = 1.5 \times (\text{Number of logged signals} + 1) \times 8 \times (\text{Logging duration(s)}) / (\text{Decimation factor} \times \text{Sample time(s)})$$

Figure 19: File Size Limit Equation

- Run the simulation according to instructions in section 5. Keep in mind the length of the simulation, so the file limit does not get exceeded.
- After the simulation is complete, reset the model to transfer the output file to the host PC.
- The output file can be retrieved from the following file path: `\models\PROJECT\SM_NAME \OpREDHAWKtarget\ .mat` where SM.NAME and project are replaced by the SM title and project names. **IF THE BUFFER OR FILE LIMIT WAS CALCULATED INCORRECTLY THE OUTPUT FILE WILL BE EMPTY.**

More info on the OpWriteFile block is available here: [website](#)

9 ePHASORSIM

To incorporate a grid model from PSS/E into the simulation, the ePHASORSIM software is used. The steps for doing this are described below.

- Confirm that dynamic stability can run properly on the case in PSS/E. Use the .raw file and .dyr file that will be used in the HIL simulation and perform a dynamic stability analysis in PSS/E. Confirm that there are no significant errors.
- Save the .raw file as **PSS/E Version 32**. At this time, only Version 32 is compatible with ePHASORSIM.
- Save the .dyr file.
- Create an excel sheet specifying the desired inputs and outputs to the ePHASORSIM block. For example, a bus fault would be an input and a bus voltage to monitor would be an output. Follow the syntax provided in **OPAL-RT's I/O Guide**. An example is shown in Figure 20.

	A	B	C	D	E
1	outgoing	Bus_voltage	bus_101/Vmag	bus_102/Vmag	bus_3008/Vmag
2	incoming	Fault	bus_101/active3PGFault	bus_102/active3PGFault	bus_3008/active3PGFault
3					
4					
5					
6					
7					

Figure 20: Excel Inputs/Outputs File

- Upload the .raw file (version 32), .dyr file, and .xlsx file to the host PC.
- Include the .raw, .dyr, and .xlsx file in the project folder → “Models” → folder with the model name.
- In the RT-LAB project workspace click “Edit” to open the Simulink model.
- In the library browser (under the simulation tab) search “ePHASORSIM” and click ePHASORSIM → Solver, then and drag and drop the block into the SM subsystem in the model.
- Double-click the solver block to open the mask.
- Click “Input file format” to open the dropdown menu and select “PSSE Version 32.”
- Select the .raw, .dyr, and .xlsx file you have included in the project folder as shown in Figure 21. Click “Apply.”
- The ePHASORSIM solver is now ready to be connected into the HIL simulation as desired. Note that the inputs and outputs specified in the Excel will be “muxed” if they are included in the same row; for instance, the three bus voltages will appear to be one signal, but a “de-mux” block in Simulink can recover the three separate signals.

The ePHASORSIM block performs a very similar function as PSS/E dynamic stability analysis, but it does so in real time. One note is to be careful with how loads are modeled – if loads are ‘converted’ from constant MVA to constant current/impedance in PSS/E before dynamic stability analysis is performed, the case must be saved *with the loads converted* before it is uploaded to ePHASORSIM to see matching results. The ePHASORSIM solver will run the dynamic stability analysis with the loads modeled however they were modeled in the .raw file (it looks for non-zero columns to determine this).

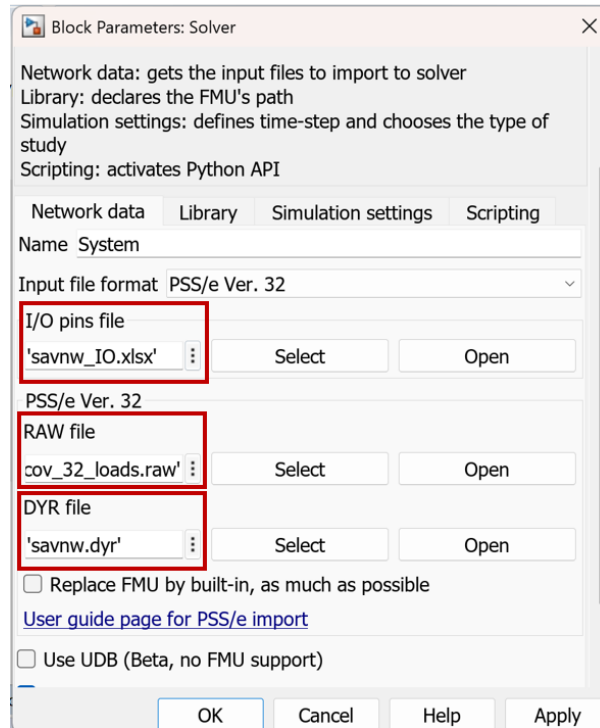


Figure 21: ePHASORSIM Parameter Popup in RT-Lab

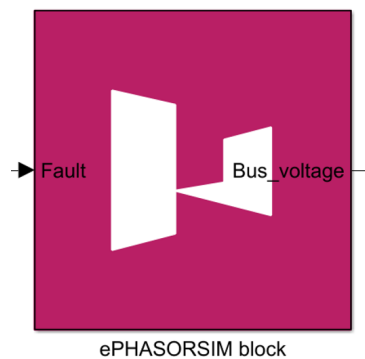


Figure 22: ePHASORSIM Block in RT-Lab

10 OPAL-RT Issues

10.1 Documentation

OPAL-RT's documentation for RT-LAB and ePHASORSIM includes:

- **Support knowledge base**, which allows users to search specific questions, but returns articles with varying degrees of usefulness.
- **RT-Lab User Guide**, which is quite in-depth but sometimes tricky to navigate/overwhelming.
- **ePHASORSIM User Guide**, which is helpful but a bit scattered; sometimes you have to dig deep to find an answer to a question.
- **OPAL-RT Support Ticket Page**, where you can submit a help ticket and work with someone from OPAL-RT to address whatever issue you are facing. This is generally very helpful, but sometimes a slow/tedious process to work through. Include as much information as you can in the ticket to streamline the process.

10.2 Random Errors

When running RT-LAB you may experience various errors. Common issues include:

- Project stuck building.
- Unable to locate OPAL-RT Board when loading.
- "RT-LAB not Installed" in RT-LAB SIMULINK blocks.

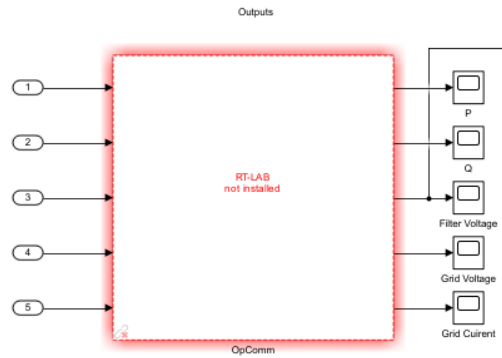


Figure 23: RT-LAB not installed error

- Program crashes/gets stuck.
- Random warnings that don't go away.

The debug loop below shows the order of methods that were used when the above errors occurred.

- Refresh RT Server (Right click → Refresh).
- Close/Reopen Project (Right click → Close Project).
- Close/Reopen RT Lab (Use task manager if necessary).
- Power off/on Host PC.
- Power on/off OPAL-RT and I/O Expansion Unit.
- Submit help ticket to OPAL-RT. Link: [OPAL-RT Support](#)

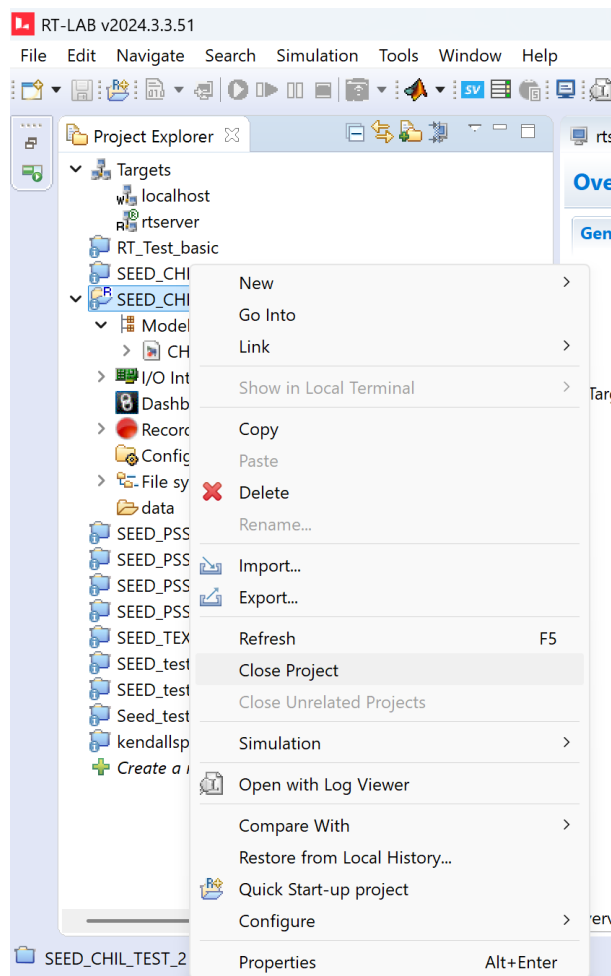


Figure 24: Close Project